

Multimedia Appendix 1

Detailed audit methods and record-level supporting tables

This appendix provides the prespecified search set, analytic definitions, aggregate sensitivity analyses, and record-level audit trails for the strict primary cohort. It contains no participant-level data.

Contents

Tables S1–S15 report search, screening, descriptive, temporal, denominator, and sensitivity analyses. Table S16 is the analytic codebook. Table S17 lists all 263 NCT identifiers in the strict primary cohort. Tables S18–S21 provide record-level audits of prospective-design reclassification, randomized-design reclassification, downstream clinical endpoints, and completed-study result reporting.

Supplementary Methods

Registry source and search

ClinicalTrials.gov records were obtained from the Aggregate Analysis of ClinicalTrials.gov database. Sixteen prespecified AI, oncology, and imaging queries were run, and records were deduplicated by NCT identifier. The locked extraction date was 2026-05-11. The search produced 846 candidate records.

Eligibility and strict primary cohort

Records were eligible for the strict primary cohort when they concerned cancer, used radiological imaging, and treated AI, machine learning, deep learning, radiomics, computer-aided detection/diagnosis, or automated image analysis as an explicit primary or major study objective. Pathology-only, endoscopy-only, genomics/EHR-only, non-oncologic, non-AI imaging, and therapeutic studies using imaging only as an outcome-assessment method were excluded. The strict primary cohort comprised 263 records with eligibility=include_core and ai_as_primary_focus=yes.

Manual adjudication and AI-specific design

Registry titles, summaries, descriptions, interventions, allocation fields, and outcomes were reviewed at the record level. Study-level prospective design indicated an interventional study or a prospectively registered observational time perspective. Prospective AI validation required a fixed AI imaging tool or workflow to be prospectively applied to newly accrued cases. Study-level randomization was based on the registered allocation field; randomized AI-workflow comparison additionally required allocation to alter exposure to AI-assisted imaging or workflow versus a standard or non-AI comparator. Table S18 identifies all 67 study-level prospective records without confirmed prospective AI validation, and Table S19 identifies all 4 study-level randomized records without confirmed AI-workflow randomization.

Endpoint hierarchy

One primary AI evaluation level was assigned from the registered main AI objective and primary outcomes: technical performance, diagnostic accuracy, reader performance, workflow efficiency, clinical decision impact, prognostic prediction, treatment-response prediction, true patient outcome, health economic, or other/unclear. True patient outcomes represented health, morbidity, survival, recurrence, or patient-reported consequences rather than model prediction of those outcomes. The downstream clinical-impact group comprised clinical-decision, true patient-outcome, or health-economic primary evaluations.

Result reporting and publication linkage

Registry result posting was determined from ClinicalTrials.gov at extraction. Completed studies were also searched in PubMed by exact NCT identifier; linked reports were counted only when they presented empirical AI results from the registered study. Exact-identifier linkage prioritizes specificity but can miss publications that omit the NCT identifier. A maturity denominator included studies completed at least 12 months before extraction.

Statistical analysis

Categorical data are summarized as counts and percentages and continuous enrollment as median and interquartile range. Exact binomial 95% confidence intervals were calculated for proportions. Exploratory temporal trends used univariable logistic regression with submitted year as a continuous variable; prespecified period comparisons (2015–2020 vs 2021–2026) used Fisher exact tests. Analyses were descriptive, two-sided, and not adjusted for multiple comparisons. Registrations in 2026 were incomplete at extraction.

Table S1. Prespecified ClinicalTrials.gov search queries

Query ID	Query term
ai_cancer_imaging	artificial intelligence cancer imaging
ml_cancer_imaging	machine learning cancer imaging
dl_cancer_imaging	deep learning cancer imaging
radiomics_cancer	radiomics cancer
radiogenomics_cancer	radiogenomics cancer
cadx_cancer_imaging	computer aided diagnosis cancer imaging
caddet_cancer_imaging	computer aided detection cancer imaging
artificial_intelligence_oncology_imaging	artificial intelligence oncology imaging
ai_oncology_radiology	artificial intelligence oncology radiology
ml_oncology_radiology	machine learning oncology radiology
ai_neoplasm_imaging	artificial intelligence neoplasm imaging
ml_neoplasm_imaging	machine learning neoplasm imaging
dl_neoplasm_imaging	deep learning neoplasm imaging
ai_lung_ct	artificial intelligence lung cancer CT
ai_breast_mammography	artificial intelligence breast cancer mammography
ai_prostate_mri	artificial intelligence prostate cancer MRI

Note: Queries were run against the locked ClinicalTrials.gov/AACCT snapshot and deduplicated by NCT identifier.

Table S2. Registry-record screening flow

Stage	Screening stage	No.
1	Candidate records identified and deduplicated	846
2	Excluded after registry-record screening	180
3	Broad or ancillary AI-imaging records retained for sensitivity analyses	403
4	Core eligible records	263
5	Primary cohort: core eligible and AI as primary focus	263

Note: The strict primary cohort required final core eligibility and AI imaging as an explicit primary or major study focus.

Table S3. AI-specific validation outcomes by registration period

Outcome	2015–2020	2021–2026
AI-specific prospective design	15/52 (28.8%; 95% CI, 18.3–42.3)	88/211 (41.7%; 95% CI, 35.3–48.4)
AI-workflow randomization	5/52 (9.6%; 95% CI, 4.2–20.6)	23/211 (10.9%; 95% CI, 7.4–15.8)
Decision, patient, or economic primary evaluation	0/52 (0.0%; 95% CI, 0.0–6.9)	14/211 (6.6%; 95% CI, 4.0–10.8)
True patient outcome endpoint	0/52 (0.0%; 95% CI, 0.0–6.9)	7/211 (3.3%; 95% CI, 1.6–6.7)
Registry results posted	3/52 (5.8%; 95% CI, 2.0–15.6)	0/211 (0.0%; 95% CI, 0.0–1.8)

Note: Percentages use all primary-cohort studies registered in each period as denominators. Confidence intervals are exact binomial intervals.

Table S4. Exploratory temporal trend models

Outcome	Annual OR (95% CI)	Annual P	2021–2026 vs 2015–2020 OR (95% CI)	Fisher P
AI-specific prospective design	1.10 (0.98–1.24)	0.096	1.70 (0.84–3.57)	0.147
AI-workflow randomization	1.08 (0.90–1.29)	0.404	1.29 (0.45–4.56)	0.808
Decision, patient, or economic primary evaluation	1.27 (0.93–1.75)	0.131	∞ (0.85–∞)	0.076

Note: Annual odds ratios were estimated with univariable logistic regression using submitted year as a continuous variable. Period comparisons used Fisher exact tests. Analyses were descriptive and were not adjusted for multiple testing.

Table S5. Primary AI evaluation levels in the strict primary cohort

Primary AI evaluation level	No.	%
Diagnostic accuracy	154	58.6%
Prognostic prediction	34	12.9%
Reader performance	27	10.3%
Technical performance	12	4.6%
Workflow efficiency	10	3.8%
Clinical decision impact	7	2.7%
Treatment response prediction	7	2.7%
True patient outcome	7	2.7%
Other unclear	5	1.9%

Note: Each registered study was assigned one primary AI evaluation level after review of its stated main AI objective and registered primary outcomes.

Table S6. Cancer types represented in the strict primary cohort

Cancer type	No.
Lung	57
Breast	54
Liver	29
Prostate	29
Brain/CNS	19
Colorectal	18
Pancreas	15
Urologic nonprostate	15
Gynecologic	14
Head neck	11
Hematologic	3

Note: Cancer categories were not mutually exclusive; therefore, counts can exceed the cohort size.

Table S7. Imaging modalities and imaging-related sources

Imaging modality/source	No.
CT	132
MRI	92
Radiomics feature imaging	51
Ultrasound	49
Mammography tomosynthesis	27
PET	25
Radiology report	18
Xray radiograph	18
Image guided procedure	13

Note: Categories were not mutually exclusive. Radiomics-feature imaging and radiology reports were retained as imaging-derived sources rather than physical modalities.

Table S8. AI method categories

AI method	No.
Automated image analysis	101
Deep learning	96
Machine learning	62
Ai workflow decision support	53
Radiomics	51
Computer aided detection diagnosis	29

Note: AI-method categories were not mutually exclusive.

Table S9. Imaging modality/source by AI method

Imaging modality/source	Ai workflow decision support	Automated image analysis	Computer aided detection diagnosis	Deep learning	Machine learning	Radiomics
CT	23	49	13	53	31	28
Image guided procedure	5	4	1	3	5	2
Mammography tomosynthesis	10	12	5	6	4	0
MRI	13	33	6	29	24	19
PET	2	7	3	6	4	5
Radiology report	5	6	4	10	3	1
Radiomics feature imaging	4	17	1	15	21	51
Ultrasound	12	22	6	17	14	5
Xray radiograph	7	7	1	7	4	4

Note: Cells show numbers of registered studies. Both dimensions were non-mutually exclusive.

Table S10. Registry result-posting estimates under alternative denominators

Denominator	Estimate
Registry results posted: all primary-cohort studies	3/263 (1.1%; 95% CI, 0.4–3.3)
Registry results posted: completed studies	3/72 (4.2%; 95% CI, 1.4–11.5)
Registry results posted: completed at least 12 months before extraction	3/60 (5.0%; 95% CI, 1.7–13.7)
PubMed-linked empirical AI results article by extraction: completed studies	13/72 (18.1%; 95% CI, 10.9–28.5)
PubMed-linked empirical AI results article by extraction: completed at least 12 months before extraction	12/60 (20.0%; 95% CI, 11.8–31.8)
Any registry-posted or PubMed-linked results by extraction: completed studies	15/72 (20.8%; 95% CI, 13.1–31.6)
Any registry-posted or PubMed-linked results by extraction: completed at least 12 months before extraction	14/60 (23.3%; 95% CI, 14.4–35.4)

Note: Posted registry results refer only to results available on ClinicalTrials.gov at extraction and do not imply peer-reviewed publication.

Table S11. Cohort-definition sensitivity analysis

Cohort	N	Prospective AI	Randomized AI workflow	Decision/patient/economic	True patient outcome	Registry results
Primary cohort	263	103 (39.2%)	28 (10.6%)	14 (5.3%)	7 (2.7%)	3 (1.1%)
Final include-core cohort regardless of AI-primary flag	263	103 (39.2%)	28 (10.6%)	14 (5.3%)	7 (2.7%)	3 (1.1%)
Final core-plus-broad cohort with AI-primary focus	450	165 (36.7%)	28 (6.2%)	79 (17.6%)	7 (1.6%)	6 (1.3%)

Note: Values are No. (%) unless otherwise indicated. The broad cohort includes ancillary or less explicit AI-imaging evaluations and is not the primary inferential cohort.

Table S12. Reclassification from study-level design to AI-specific design

Design domain	Study-level positive	AI-specific yes	AI-specific no	AI-specific unclear	AI-specific confirmation
Prospective design	170	103	53	14	60.6%
Randomized design	32	28	3	1	87.5%

Note: AI-specific confirmation is calculated among studies positive on the corresponding study-level registry design field.

Table S13. Stage-appropriate denominators for clinical evaluation

Evidence stage	N	Prospective AI	Randomized AI workflow	Completed	Unknown status
Human-interaction or downstream-effect evaluation	51	33	11	13	12
Decision, patient, or economic primary evaluation	14	14	6	3	3

Note: These nested evidence-stage groups prevent technical or diagnostic studies from being interpreted as failed patient-outcome evaluations.

Table S14. Scope sensitivity analyses

Cohort	N	Prospective AI	Randomized AI workflow	Decision/patient/economic	True patient outcome
Full primary cohort	263	103 (39.2%)	28 (10.6%)	14 (5.3%)	7 (2.7%)
Excluding radiomics-only studies	244	101 (41.4%)	28 (11.5%)	14 (5.7%)	7 (2.9%)
Excluding radiology-report-only studies	261	101 (38.7%)	27 (10.3%)	13 (5.0%)	7 (2.7%)
Excluding radiomics-only and radiology-report-only studies	242	99 (40.9%)	27 (11.2%)	13 (5.4%)	7 (2.9%)

Note: Values are No. (%) unless otherwise indicated.

Table S15. Validation markers by registry status

Registry status	N	Prospective AI	Randomized AI workflow	Decision/patient/economic	True patient outcome
Completed	72	22	9	3	2
Ongoing or other	126	59	15	8	3
Unknown status	65	22	4	3	2

Note: Registry status was grouped as completed, ongoing/other, or unknown at extraction.

Table S16. Analytic codebook

Variable	Definition	Allowed values	Analysis role
eligibility	Final eligibility based on oncologic context radiological imaging and whether AI imaging was the principal object of evaluation	include_core;include_broad;exclude	Primary cohort and sensitivity analysis
ai_as_primary_focus	Whether AI imaging was an explicit primary or major objective intervention comparator or prespecified evaluation	yes;no;unclear	Primary cohort definition
adjudication_note	Record-specific rationale supporting final eligibility and AI-focus classification	free text	Audit trail
study_level_prospective_design	Whether the registered study was interventional or an observational study with prospective time perspective	yes;no;unclear	Paired design reclassification
prospective_ai_validation	Whether a fixed AI imaging tool or workflow was prospectively applied to newly accrued cases	yes;no;unclear	AI-specific design
study_level_randomized_design	Whether the registered allocation field indicated randomization	yes;no;unclear	Paired design reclassification
randomized_ai_workflow_comparison	Whether allocation changed exposure to AI-assisted imaging or workflow versus standard or non-AI care	yes;no;unclear	AI-specific design
primary_ai_evaluation_level	Single principal AI-specific evaluation level assigned from registered primary outcomes and the explicit main AI objective	technical_performance;diagnostic_accuracy;reader_performance;workflow_efficiency;clinical_decision_impact;prognostic_prediction;treatment_response_prediction>true_patient_outcome;health_economic;other_unclear	Endpoint position
true_patient_outcome	Whether the registered primary AI evaluation concerned patient health or a patient-reported consequence of AI use	yes;no;unclear	Patient-centered endpoint
human_or_downstream_evaluation	Whether the primary level was reader workflow decision patient or economic evaluation	yes;no	Stage-appropriate denominator
downstream_clinical_impact_endpoint	Whether the primary level was clinical decision impact true patient outcome or health-economic evaluation	yes;no	Downstream endpoint denominator
radiomics_only	Whether radiomics was the only recorded AI-method category	yes;no	Scope sensitivity
radiology_report_only	Whether radiology reports were the only recorded imaging source	yes;no	Scope sensitivity
registry_results_posted	Whether ClinicalTrials.gov indicated posted results at extraction	true,false	Result linkage
empirical_results_article_linked_by_exact_nct	Whether a completed study had an empirical AI-results PubMed article linked by exact NCT identifier by extraction	true,false;blank	Publication linkage
submitted_year	Calendar year derived from first-submitted date	2015-2026	Temporal analysis
registration_period	Prespecified descriptive submission period	2015-2020;2021-2026	Stratified analysis
cancer_types	Semicolon-delimited non-mutually-exclusive cancer categories	registry-derived categories	Landscape analysis
imaging_modalities	Semicolon-delimited non-mutually-exclusive imaging modalities or imaging-related sources	registry-derived categories	Landscape analysis
ai_method_categories	Semicolon-delimited non-mutually-exclusive AI-method labels	registry-derived categories	Landscape analysis
imaging_tasks	Semicolon-delimited clinical or technical imaging tasks	registry-derived categories	Landscape analysis
center_category	Classification from registered facility count	single_center;multicenter;not_reported	Study characteristics
extraction_date	Date of locked registry snapshot	2026-05-11	Version control

Note: Allowed values are shown as semicolon-delimited categories. Free-text registry fields are retained in Multimedia Appendix 2.

Table S17. NCT identifiers in the strict primary cohort

NCT identifier 1	NCT identifier 2	NCT identifier 3	NCT identifier 4
NCT02440139	NCT02505698	NCT02745600	NCT02757846
NCT02943824	NCT03064828	NCT03151564	NCT03179150
NCT03198975	NCT03237130	NCT03487952	NCT03552497
NCT03687918	NCT03706534	NCT03708978	NCT03750890
NCT03753724	NCT03787667	NCT03829423	NCT03851497
NCT03857373	NCT03887598	NCT03981848	NCT04022512
NCT04027556	NCT04036903	NCT04060706	NCT04154228
NCT04156880	NCT04164186	NCT04191980	NCT04214782
NCT04215211	NCT04217018	NCT04270032	NCT04271657
NCT04273451	NCT04313140	NCT04317313	NCT04340180
NCT04394052	NCT04407039	NCT04409340	NCT04441775
NCT04454450	NCT04482114	NCT04497233	NCT04511481
NCT04527510	NCT04535466	NCT04635488	NCT04688086
NCT04732156	NCT04765150	NCT04778670	NCT04816981
NCT04832594	NCT04834596	NCT04837833	NCT04838756
NCT04843176	NCT04893200	NCT04894955	NCT04949776
NCT04954378	NCT04996615	NCT05021055	NCT05024591
NCT05048095	NCT05096533	NCT05102240	NCT05126160
NCT05161624	NCT05170282	NCT05230563	NCT05231616
NCT05243121	NCT05257694	NCT05324046	NCT05380609
NCT05385718	NCT05389774	NCT05425134	NCT05443412
NCT05443672	NCT05481762	NCT05513638	NCT05523245
NCT05541042	NCT05542992	NCT05547971	NCT05589480
NCT05615298	NCT05617469	NCT05627297	NCT05640011
NCT05653063	NCT05656053	NCT05658679	NCT05689437
NCT05698056	NCT05704920	NCT05723965	NCT05732974
NCT05739331	NCT05739500	NCT05769517	NCT05775068
NCT05802186	NCT05804799	NCT05817110	NCT05820113
NCT05826197	NCT05858762	NCT05873972	NCT05892380
NCT05925738	NCT05960188	NCT05968157	NCT05979883
NCT06032390	NCT06037343	NCT06044454	NCT06056999
NCT06088134	NCT06092450	NCT06104189	NCT06163846
NCT06168864	NCT06171607	NCT06187350	NCT06200116
NCT06224283	NCT06267690	NCT06281769	NCT06282068
NCT06285058	NCT06295497	NCT06298305	NCT06306599
NCT06314750	NCT06323408	NCT06331130	NCT06331975
NCT06336694	NCT06342934	NCT06356441	NCT06360523
NCT06362291	NCT06365905	NCT06366906	NCT06392737
NCT06423066	NCT06428799	NCT06441292	NCT06451393

NCT identifier 1	NCT identifier 2	NCT identifier 3	NCT identifier 4
NCT06463392	NCT06468332	NCT06479772	NCT06496360
NCT06498674	NCT06510127	NCT06521788	NCT06528223
NCT06528236	NCT06531343	NCT06540196	NCT06541288
NCT06546072	NCT06559046	NCT06557361	NCT06576232
NCT06578988	NCT06589154	NCT06604078	NCT06604689
NCT06622967	NCT06626087	NCT06626178	NCT06632886
NCT06637059	NCT06638398	NCT06638866	NCT06641947
NCT06643715	NCT06657027	NCT06659601	NCT06684418
NCT06703112	NCT06737367	NCT06738459	NCT06741423
NCT06746324	NCT06747299	NCT06757530	NCT06761742
NCT06773598	NCT06804785	NCT06825078	NCT06842264
NCT06853457	NCT06859840	NCT06885333	NCT06888089
NCT06910956	NCT06930586	NCT06932133	NCT06934239
NCT06945692	NCT06947096	NCT06964412	NCT06984289
NCT07005791	NCT07020169	NCT07040527	NCT07050576
NCT07056270	NCT07065422	NCT07075679	NCT07084779
NCT07088354	NCT07107035	NCT07111364	NCT07117045
NCT07117786	NCT07121972	NCT07153783	NCT07162194
NCT07166445	NCT07181954	NCT07189520	NCT07190040
NCT07231627	NCT07235410	NCT07251946	NCT07256457
NCT07269535	NCT07280559	NCT07282184	NCT07287111
NCT07304492	NCT07332923	NCT07350382	NCT07360145
NCT07373951	NCT07376304	NCT07401199	NCT07406958
NCT07408531	NCT07411443	NCT07427407	NCT07439757
NCT07458425	NCT07462351	NCT07463833	NCT07491055
NCT07495735	NCT07496684	NCT07509632	NCT07532083
NCT07538882	NCT07552532	NCT07561814	

Note: All 263 identifiers are unique. Hyperlinks lead to the corresponding ClinicalTrials.gov record.

Table S18. Study-level prospective records without confirmed prospective AI validation

NCT identifier	Registered title	AI-specific classification	Registry text excerpt (truncated)	Review basis
NCT02440139	Evaluation of Computer-Aided Lung Nodule Detection Software in Thoracic CT for Riverain Technologies LLC	No	STUDY DESIGN: This is a retrospective, multi-reader, multi-case, (MRMC) randomized reader study.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT02505698	A Data Collection Study for the Evaluation of a Novel Infra-red Breast Imaging System for Risk Assessment	No	Three-dimensional functional Metabolic Imaging (3D MIRA) is a new infrared imaging technology using the Real Imager 8 (RI8) developed by Real Imaging.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT02745600	Clinical Intervention Modelling, Planning and Proof for Ablation Cancer Treatment	No	The main objective of the project is to bring the existing radio frequency ablation (RFA) model for liver cancer treatment (Project IMPACT, Grant No.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT03237130	Prediction of Lymphatic Metastasis in Esophageal Cancer by CT Radiomics	No	This study proposes to establish a CT radiomics-based prediction model for identifying metastasis lymph nodes in esophageal cancer.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT03750890	Visual Study of Molecular Genotype in Glioma Evolution	No	Based on the previous research, this project intends to establish a multi-dimensional database of glioma from clinical, radiomics and microomics levels.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT03753724	IDEAL: Artificial Intelligence and Big Data for Early Lung Cancer Diagnosis Study	Unclear	This study aims to test the use of novel CT image analysis techniques to enable a better characterisation of small pulmonary nodules.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT03787667	Application of Radiomics in Precise Preoperative Diagnosis and Prognosis Evaluation of Colorectal Cancer.	Unclear	This is a prospective, clinical study.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT03829423	An Enhanced Artificial Intelligence Breast MRI Interpretation System	No	This project aims to develop a technical solution that addresses this healthcare challenge by developing a system that is able to automatically interpret breast MR images in order to aid the radiologist in their diagnosis.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT03851497	Application of Deep-learning and Ultrasound Elastography in Opportunistic Screening of Breast Cancer	Unclear	This multi-center study intends to further validate the diagnostic efficiency of S-detect and E-breast in opportunistic breast cancer screening populations in China.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT03857373	Renal Cancer Detection Using Convolutional Neural Networks	No	We aim to experiment and implement various deep learning architectures in order to achieve human-level accuracy in Computer-aided diagnosis (CAD) systems.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04060706	Heuristics, Algorithms and Machine Learning: Evaluation & Testing in Radiation Therapy	No	Developing machine learning algorithms for autosegmentation of normal tissue anatomy, and to extend machine learning algorithms to identify and segment normal tissue structures in cone beam CT images, and to utilise the ML segmentations to evaluate image signatures correlated with treatment toxicity	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04154228	Non-invasive Evaluation of Lymphoma Patients Based on Artificial Intelligence and PET/MRI	No	The aim of this prospective study is to develop an automatic artificial intelligence-based tool for the assessment of early response to treatment and evaluation of residual masses in patients with lymphoma.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04313140	Virtual Biopsy Development to Identify Patient's Glioma Grade at the CHU of Poitiers Using Magnetic Resonance Imaging	No	Glioma is a tumor of the central nervous system.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04394052	Novel Imaging Techniques for the Characterization of Musculoskeletal Tumors II	Unclear	DIAGNOSTIC_TEST MR imaging Medical imaging Lesion benignancy or malignancy Histologic determination of lesion aggressiveness (benign versus malignant) on core biopsy material secondary Sarcoma FNCLCC (fédération Nationale des Centres de Lutte Contre le Cancer) grade Histologic grade of the sarcomas...	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT04407039	Establishment and Evaluation of Multimodal Image Recognition System of Glioma Based on Deep Learning	No	With the continuous development of basic and clinical research of glioma and the advent of various new drugs and treatment technologies, molecular pathological diagnosis based on the individual level of glioma patients is particularly important.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04454450	Using Advanced Imaging Studies to Develop a Profile of High-grade Serous Ovarian Cancer	No	The researchers are doing this study to find out whether the researchers can combine information provided by PET/MRI scans with information from tests on blood and tissue samples to develop a very detailed description (profile) of high-grade serous ovarian carcinoma (HGSOc), which could improve our...	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04535466	Diagnosis Predictive Model for Dense Density Breast Tissue Based on Radiomics	No	It is a prospective, observational cohort study of patients with dense breast tissue.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04635488	Prediction Model for Lateral Lymph Node Metastasis	No	Then a part of the nodes would be used as the training set to build the prediction model, and the another part nodes would be used as the validation set.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.

NCT identifier	Registered title	AI-specific classification	Registry text excerpt (truncated)	Review basis
NCT04732156	Detection of ISUP≥2 Prostate Cancers Using Multiparametric MRI: Prospective Multicenter Comparison of the PI-RADS Score and an Artificial Intelligence System	Unclear	However, one of the weaknesses of artificial intelligence systems is their low robustness when tested on MRI images from different manufacturers or institutions.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT04816981	AI-EBUS-Elastography for LN Staging	No	The algorithm will be trained to read the images first, and then predict whether these images show cancer in the lymph node.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT04894955	Micrima MARIA Data Collection for Machine Learning Study	Unclear	The use of artificial intelligence will provide additional, novel functionality to the MARIA® system but requires participant data in order to develop and validate the machine learning algorithms that aim to increase the accuracy and overall clinical utility of the device.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT04996615	Artificial Intelligence Analysis for Magnetic Resonance Imaging in Screening Breast Cancer in High-risk Women	Unclear	Use Convolutional Neural Networks Analysis for Classification of Contrast-enhancing Lesions at Multiparametric Breast MRI.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05021055	Multicentric Study for External Validation of a Deep Learning Model for Mammographic Breast Density Categorization	Unclear	Once a system designed to aid healthcare professional decision making is developed, it must be validated.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05126160	Immune Score Based Radiomics in Nasopharyngeal Carcinoma	No	In nasopharyngeal carcinoma (NPC), tumor, node and metastasis (TNM) staging system is the main tool for determining treatment strategy and assessment of prognosis.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05230563	A Clinical Study of LUNG-CT CAD Decision Support	No	Users can use existing medical image capture and transmission systems or reading software to view these annotations on the workstation.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05231616	Deep Learning Model for Diagnosis and Contour of Cervical Lymph Node for Nasopharyngeal Carcinoma	No	Thus, the investigators aimed to develop and externally validate a deep learning model to assist in the diagnosis and localization of metastatic lymph nodes in nasopharyngeal carcinoma.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05243121	Artificial Intelligence Analysis for Magnetic Resonance Imaging in Screening and Diagnosis of Breast Cancer	Unclear	DIAGNOSTIC_TEST MRI undergoing enhanced MRI Breast Cancer Screening Compare the area under the curve of the deep learning model of the BMRI full sequence, contrast-enhanced and non-contrast-enhanced sequence in the diagnosis of breast cancer.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05425134	Deep Learning Signature for Predicting Occult Nodal Metastasis of Clinical N0 Lung Cancer	Unclear	The purpose of this study is to evaluate the performance of a PET/CT-based deep learning signature for predicting occult nodal metastasis of clinical stage N0 non-small cell lung cancer in a multicenter prospective cohort.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05443412	Artificial Intelligence (AI)-Assisted Risk-based Prostate Cancer Detection	No	This is a prospective clinical study recruiting 510 men at risk of PCa to undergo urine, blood, AI-assisted ultrasound and AI-assisted MRI investigations to stratify risk of clinically significant PCa (csPCa).	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05523245	Predicting the Efficacy of Neoadjuvant Therapy in Patients With Locally Advanced Rectal Cancer Using an AI Platform Based on Multi-parametric MRI	No	Establish a deep learning model based on multi-parameter magnetic resonance imaging to predict the efficacy of neoadjuvant therapy for locally advanced rectal cancer. This study intends to combine DCE with conventional MRI images for DL, establish a multi-parameter MRI model for predicting the efficacy...	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05541042	Radiogenomics in Glioblastoma: Correlation Between Multiparametric Imaging Biomarkers and Genetic Biomarkers	No	The purpose of this study is to evaluate relationships between multiparametric imaging biomarkers and genetic analysis in glioblastoma patients.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05542992	Deep Learning Model for Pure Solid Nodules Classification	Unclear	The purpose of this study is to compare the predictive performance of a CT-based deep learning model for pure-solid nodules classification and compared with the tumor maximum standardized uptake value on PET in a multicenter prospective cohort.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05547971	Development of Intelligent Model for Radioactive Brain Damage of Nasopharyngeal Carcinoma Based on Radio-metabolomics	No	This project focuses on the early prediction and diagnosis of radiation-induced brain injury in nasopharyngeal carcinoma patients.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05656053	Intraoperative Rapid Diagnosis of Glioma Based on Fusion of Magnetic Resonance and Ultrasound Imaging	No	Their imaging data, genotype data, clinical history data, and pathology data will be used for the experimental study.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05732974	A Machine Learning Approach to Identify Patients With Resected Non-small-cell Lung Cancer With High Risk of Relapse	Unclear	There are currently no validated prognostic biomarkers able to identify patients with a high risk of relapse.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05826197	Prediction of Axillary Lymph Node Metastasis Status in Breast Cancer Based on PET/CT Radiomics	No	Patients with suspected breast cancer undergoing PET/CT at our hospital.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.

NCT identifier	Registered title	AI-specific classification	Registry text excerpt (truncated)	Review basis
NCT05858762	Artificial Intelligence for Automated Diagnosis of Breast Cancer	No	The study is a prospective observational study based on Artificial intelligence whose the mail goal is to develop a method to identify a lesion.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT05892380	The Clinical Value of an Artificial Intelligence System Using Abbreviated Protocol of Breast MRI Facilitates Classification of Breast Lessons	Unclear	This study aims to use a combination of abbreviated protocol and artificial intelligence to automatically identify lesions and make diagnosis without decreasing the diagnostic accuracy of breast cancer, thus enhancing the comfort of patient examination, accelerating the flow of examination and redu...	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.
NCT05960188	Project 1: Self-Triage by 2D Full-field Digital Mammography or Synthetic Images	No	One method of breast cancer screening involves radiologists reading digital tomosynthesis (DBT) images.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06056999	New Method to Differentiate Benign and Malignant Pulmonary Nodules.	No	Peripheral blood of subjects will be collected and detected by mass spectrometry to obtain the metabolomic characterization.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06104189	Prospective Ultrafast MRI Radiomics for Breast Cancer	No	The goal of this prospective study is to investigate the performance of ultrafast MRI radiomics in classifying histological factors and subtypes of breast cancer compared with standard MRI among radiologists with varying experience.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06163846	Image Mining and ctDNA to Improve Risk Stratification and Outcome Prediction in NSCLC Applying Artificial Intelligence.	No	Secondly, image mining approach has been proposed to provide prognostic information at baseline evaluation, as also shown in our previous work.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06171607	Contrast Enhanced Ultrasound Medical Imaging for Identifying Breast Masses	No	To develop a CEUS-based radiomics workflow to extract radiomic metrics (> 1600 features) in classifying breast mass malignancy (Radiomics).	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06298305	Artificial Intelligence & Prostate Cancer	No	This is a single center, blinded (the mpMRI will be read by expert personnel not aware of the patient's diagnosis), retrospective study.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06323408	Integrated Analysis of Therapy Response and Resistance in Embryonal Tumors and Gliomas	No	The tumor biology and clinical needs to diagnose and treat these tumors are comparable across all age groups, so an integrated treatment environment overseen by adult and pediatric neuro-oncology specialists seems promising to leverage synergisms and advance diagnostic and therapeutic development i...	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06331130	Mesenteric Infiltration in Ovarian Cancer	No	To evaluate if CT features at diagnosis in patients with HGSOc can be used to build an Artificial Intelligence model capable of discerning the pathological involvement of the mesentery, assessing the potential impediments for an optimal debulking surgery and predicting the development of resistance...	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06423066	Developing a Machine Learning Model to Predict Pleural Adhesion Preoperatively Using Pleural Ultrasound	No	With the advancement of ultrasound technology and the need to address clinical challenges, ultrasound imaging technology has seen new developments in the field of thoracoscopic surgery.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06468332	Artificial Intelligence-based Platform, Integrating Pathologic, Imaging and Molecular Profiles of Prostate Cancer	No	A retrospective cohort of 200 radical prostatectomy patients will be identified within three participating clinical centres.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06531343	Advancing Lung Cancer Screening: Artificial Intelligence, Multimodal Imaging and Cutting-Edge Technologies for Early Detection and Characterization	No	This project aims to develop a new approach exploiting both these revolutionary advancements to bridge the existing gap in lung cancer screening.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06589154	The Application of Multimodal Artificial Intelligence Systems in Prostate Cancer Diagnosis and Prognosis Analysis	No	This multi-center study aims to develop and validate a non-invasive, multi-modal artificial intelligence model that combines cell-free DNA (cfDNA) profiles with multi-parametric MRI (mpMRI).	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06604689	AI-guided Prognostication and Cranial Radiotherapy Optimization in EGFR-TKI-treated Non-small Cell Lung Cancer Patients With Baseline Brain Metastases	No	The goal of this observational study is to extract the imaging features of brain lesions and primary lung lesions in NSCLC patients with brain metastases by deep learning, as well as common clinicopathological parameters, which are used to construct a multimode model that can accurately predict the...	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06626178	Development of an Artificial Intelligence Model for the Identification and Prevention of Smoking-related Diseases.	No	The study has the main objective of training and validate a reliable and unbiased Artificial Intelligence (AI) algorithm that detects the presence of nodules and differentiates between malignant or benign tumor types.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06684418	Artificial Intelligence-based Model for the Prediction of Occult Lymph Node Metastasis and Improvement of Clinical Decision-making in Non-small Cell Lung Cancer	No	This nationwide, multicenter observational study aims to develop and validate a multimodal artificial intelligence (AI) model for detecting occult lymph node metastasis in early-stage non-small cell lung cancer (NSCLC) patients.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06757530	Radiomics Model for Assessing Lymph Node Status in cN0 Patients withHNSCC	Unclear	DIAGNOSTIC TEST AI Using artificial intelligence models to distinguish between patients with lymph node metastasis and those without lymph node metastasis.	Prospective recruitment or data collection was stated, but the registry did not establish when the AI system was locked or whether it was applied before clinical outcomes accrued.

NCT identifier	Registered title	AI-specific classification	Registry text excerpt (truncated)	Review basis
NCT06804785	Hybrid Intelligence for Trustable Diagnosis and Patient Management of Prostate Cancer (HIT-PIRADS)	No	The purpose of this study is to develop a new way to diagnose prostate cancer through the use of artificial intelligence.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06842264	The Development and Validation of MRI-AI-based Predictive Models for csPCa	No	This study retrospectively included patients who underwent prostate magnetic resonance imaging (MRI) and subsequent ultrasound-guided prostate biopsy at Peking University First Hospital from January 2019 to December 2023, and prospectively enrolls patients from January 2024 to December 2029.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06859840	LEAF(Liver Tumor dEtection And classiFication AI)	No	For cases where the model indicates malignancy without clear evidence from medical history or other data, follow-up will be performed to confirm the true value through pathological results.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06885333	Whole-Brain Tumor Burden Metrics with Inflammatory and Molecular Markers to Predict Postoperative Neurocognitive Decline in Glioma	No	This study aims to develop a model using whole-brain tumor burden metrics, inflammatory and molecular markers for predicting high risk of neurocognitive decline (ND) postoperatively.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06932133	A Trial on the Use of Point-of-care Ultrasound in the Assessment of Breast Symptoms	No	The study will be divided in two parts: a prospective interventional study and a retrospective multicase multireader study.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT06947096	Radiomics-Based AI Model for Predicting Para-Aortic Lymph Node Metastasis in Gastric Cancer Patients	No	This study aims to develop and validate an artificial intelligence (AI) model based on radiomics features extracted from preoperative CT images to predict para-aortic lymph node (PALN) metastasis in patients with gastric cancer.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07005791	AI-Based Ultrasound Prediction Model for Intracranial Pressure and Prognosis in Lung Cancer Patients With Leptomeningeal Metastasis: A Dual-Center Study	No	This study focuses on developing an innovative, artificial intelligence-based model using optic nerve sheath ultrasound videos to predict intracranial pressure in lung cancer patients with leptomeningeal metastasis.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07056270	Development of a Set of Auxiliary Decision-making System for the Perioperative Period of Hepatectomy Based on Static CT and Artificial Intelligence.	No	The study will prospectively recruit patients with chronic liver disease and liver cancer for static CT scans to establish a high-definition CT database.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07107035	Multimodal Large Model-Driven Risk and Prognosis Assessment for Brain Metastases in Lung Cancer	No	The goal of this nationwide, multicenter observational study is to develop and externally validate multimodal large models that can (1) predict the risk of brain metastases and (2) estimate long-term prognosis in patients with non-small cell lung cancer (NSCLC).	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07189520	1. SAFE-AI ONCO-TRACK: Multimodal GenAI for Early Detection of Minimal Residual Disease and Recurrence in Gastrointestinal Oncology	No	Current decision tools (TNM, MRI/PET, CEA, and other serum markers, as well as single-marker genomics) are insufficiently predictive of responders, fail to detect early MRD in many cases, and rarely connect molecular biology to dynamic perioperative data.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07256457	Postoperative Pulmonary Function Assessment Based on Deep Learning Study	No	However, most existing research is retrospective or derived from single-center data, with a primary focus on short-term safety and technical feasibility.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07287111	Effectiveness of CadAI-B Dx for Decision Support in Breast Ultrasound	No	This is a retrospective, fully-crossed, multi-reader, multi-case (MRMC) study to evaluate the effectiveness of 'CadAI-B Dx' (CadAI-B) for decision support in breast ultrasound.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.
NCT07401199	Multimodal AI for Predicting Response to Neoadjuvant Immunotherapy in Gastric Cancer (PRISM-GC)	No	The PRISM-GC study aims to develop and validate a novel Artificial Intelligence (AI) system to address this need.	Registry text did not establish prospective application of a locked AI system; the described work used retrospective/existing data, post-accrual analysis, or model development.

Note: This audit includes 53 records classified no and 14 classified unclear for prospective AI validation. Registry excerpts are truncated for display; full registry text and structured fields are provided in Multimedia Appendix 2.

Table S19. Study-level randomized records without confirmed randomized AI-workflow comparison

NCT identifier	Registered title	AI-workflow classification	Registered comparison (truncated)	Review basis
NCT03064828	Colorectal Polyps Screening With Low-dose CT Colonoscopy	No	RADIATION CT colonoscopy(normal dose) Perform CT colonoscopy with full bowel preparation and inflated colon,CT Scans on subjects with normal-dose protocol by setting scanning parameters as 120-140 kVp, 100-200mA ; RADIATION CT colonoscopy (low dose) Perform CT colonoscopy with full bowel preparation and inflated colon,CT Scans on subjects with normal-dose p...	Randomization compared CT-dose protocols, not exposure to AI assistance.
NCT04832594	Using AI to Select Women for Supplemental MRI in Breast Cancer Screening	No	OTHER AI selection for supplemental breast MRI An AI tool will generate scores used to determine eligibility. Women randomized to MRI will be examined in an MRI scanner.	AI selected eligibility within the screening strategy, but the registered randomization did not compare AI-assisted with non-AI selection or interpretation.
NCT06498674	Clinical Application of AI-assisted Ultrasound Technology in the Preoperative Evaluation of Thyroid Cancer	No	DIAGNOSTIC_TEST Supplementary Pathological Examination For the same thyroid nodule, if there is discordance between the interpretations of AI-assisted ultrasound and traditional two-dimensional ultrasound, supplementary fine needle aspiration biopsy or intraoperative biopsy will be performed to clarify the nature of the nodule. ; OTHER No Need for Supplemen...	Randomization concerned supplementary pathologic examination after discordant imaging, not exposure to an AI-assisted versus non-AI workflow.
NCT06528223	Research of the Application of Artificial Intelligence Model "PANDA"	Unclear	DIAGNOSTIC_TEST biopsy or operation To obtain a biopsy pathology or surgical pathology according to the clinical process of PDAC.	The allocation field indicated randomization, but the registered arms and AI exposure were insufficiently specified to confirm an AI-versus-non-AI comparison.

Note: The allocation field alone was not sufficient: confirmation required random assignment to change exposure to AI assistance or an AI-enabled workflow.

Table S20. Registered studies with downstream clinical-impact primary endpoints

NCT identifier	Registered title	Primary evaluation level	Registered primary endpoint (truncated)	Prospective AI	Randomized AI workflow	Registry status
NCT04832594	Using AI to Select Women for Supplemental MRI in Breast Cancer Screening	True patient outcome	Significantly Delayed Breast Cancer Detection per 1000 women Composite end-point defined by either: 1. Interval Cancer, 2. Cancer with lymph node metastasis, 3. Cancer with invasive component larger than 15 mm	Yes	No	UNKNOWN
NCT04838756	Mammography Screening With Artificial Intelligence (MASAI)	True patient outcome	Interval-cancer rate Women with interval cancer per 1000 screens	Yes	Yes	COMPLETED
NCT05389774	DOLCE: Determining the Impact of Optellum's Lung Cancer Prediction Solution	Clinical decision impact	Determine the potential effect of the LCP on discharge Measured difference between standard care, LCP and LCP-guided care for: * Percentage of cancer patients discharged (straight after assessment of the baseline scan) * Percentage of benign-nodule patients discharged (straight after assessment of the baseline scan)	Yes	No	RECRUITING
NCT05689437	MIRA Clinical Learning Environment (MIRACLE): Lung	Clinical decision impact	Rates of true positive diagnosis of ILD increase with high/low patient risk predictions being made available to clinicians. An expert review of the cases and chart review will be correlated with survey responses to determine whether the rate of true positive cases were impacted by the implementation of the MIRACLE pathways. ; Previously difficult-to-assess...	Yes	No	UNKNOWN
NCT05802186	Stereotactic Body Radiation Therapy Planning With Artificial Intelligence-Directed Dose Recommendation for Treatment of Primary or Metastatic Lung Tumors, RAD-AI Study	True patient outcome	Local failure free survival (LFS) LFS is defined as the length of time after SBRT that the patient survives without local failure (as assessed by tumor imaging). ; LFS LFS data will be collected from the end of SBRT, until the patient experiences recurrence, completes the 5-year follow-up after SBRT, or experiences death from any cause (whichever is sooner).	Yes	No	RECRUITING
NCT05873972	Multimodal Model for Efficacy Prediction Cetuximab in Colorectal Cancer Liver Metastasis Patient	True patient outcome	response rate response rate will be assessed by local MDT	Yes	Yes	UNKNOWN
NCT05979883	Randomized Evaluation of Machine Learning Assisted Radiation Treatment Planning vs Standard Radiation Treatment Planning	True patient outcome	Change in swallowing score Will be assessed using the European Organization for Research and Treatment of Cancer Quality of Life Questionnaire Head and Neck module (EORTC QLQ-H&N35). The EORTC QLQ-H&N35 is a 35-item questionnaire used to assess symptoms encountered specifically by patients with head and neck cancer. The H&N35 consists of 7 multiple-item...	Yes	Yes	COMPLETED
NCT06498674	Clinical Application of AI-assisted Ultrasound Technology in the Preoperative Evaluation of Thyroid Cancer	Clinical decision impact	Supplementary Pathological Examination For patients who have completed traditional two-dimensional color ultrasound and AI-assisted ultrasound, record whether a supplementary pathological examination was conducted. ; Change in Treatment Decision For patients who have completed traditional two-dimensional color ultrasound, AI-assisted ultrasound, and undergo...	Yes	No	RECRUITING
NCT06638398	Biomarker Platform (Virtual Nodule Clinic) for the Management of Indeterminate Pulmonary Nodules	Clinical decision impact	Appropriate nodule management Will be defined as referral for imaging surveillance for ultimately benign nodules, and biopsy, positron emission tomography scan, surgery or empiric stereotactic body radiation therapy for malignant nodules. Standard consort diagram will be created. Descriptive statistics including means, standard deviations, and median and in...	Yes	Yes	ACTIVE NOT RECRUITING
NCT06888089	Deep Clinical Trajectory Modeling to Optimize Accrual to Cancer Clinical Trials	Clinical decision impact	Percentage of Patients Enrolling in Any Dana-Farber Cancer Institute Therapeutic Clinical Trial of Anti-Cancer Systemic Therapy This measure assesses the proportion of patients in each study arm who enroll in any Dana-Farber Cancer Institute (DFCI) therapeutic clinical trial involving anti-cancer systemic therapy during the intervention period. Trial enroll...	Yes	Yes	COMPLETED
NCT06910956	Deep Learning Using Chest X-Rays to Identify High Risk Patients for Lung Cancer Screening CT	Clinical decision impact	Proportion completing Lung Cancer screening CT in 6 months after visit To assess impact on lung cancer screening CT participation (defined as completing lung cancer screening CT) in the 6 months after the baseline visit.	Yes	Yes	RECRUITING
NCT07040527	Comparative Study on Medical Artificial Intelligence Algorithm Assisted and Conventional Imaging Examination Methods	Clinical decision impact	Pathological results of surgical margins for chest wall tumors Pathological diagnosis of residual tissue cells after tumor resection	Yes	No	NOT YET RECRUITING
NCT07282184	Multimodal Deep Learning for Postoperative Liver Cancer Risk Stratification and Intervention	True patient outcome	2-year Recurrence-free Survival (RFS) Rate The primary outcome is the proportion of patients who remain free of recurrence at 2 years after curative liver resection. Recurrence is defined as radiologically confirmed (via contrast-enhanced CT or MRI according to RECIST 1.1 criteria) appearance of new lesions (intrahepatic or extrahepatic) or death from any c...	Yes	No	RECRUITING
NCT07458425	The SENTINL-1 Study: Evaluating Patient-Reported Outcomes of AI-Inferred Lung Cancer Risk	True patient outcome	Patient Reported Outcomes Measurement Information System (PROMIS) survey results before and following the return of results (ROR) To evaluate participant reported outcomes before and after return of results (ROR) using the PROMIS test surveys ; Multidimensional Impact of Cancer Risk Assessment (MICRA) following return of results (ROR) To evaluate participan...	Yes	No	RECRUITING

Note: The 14 records comprised 7 clinical-decision-impact and 7 true patient-outcome evaluations; no primary health-economic evaluation was identified.

Table S21. Completed-study result-reporting and exact-NCT publication audit

NCT identifier	Registered title	Completion date	Completed ≥12 months before extraction	Registry results posted	Empirical AI-results article linked by exact NCT	Linked PMID(s)	Any public result identified
NCT02440139	Evaluation of Computer-Aided Lung Nodule Detection Software in Thoracic CT for Riverain Technologies LLC	2018-03-31	Yes	Yes	No		Yes
NCT02943824	Machine-learning Optimization for Prostate Brachytherapy Planning	2018-09-04	Yes	No	No		No
NCT03179150	Clinical Validation of the M5L Lung CAD	2017-06-01	Yes	No	No		No
NCT03237130	Prediction of Lymphatic Metastasis in Esophageal Cancer by CT Radiomics	2020-05-15	Yes	No	No		No
NCT03708978	Development of Artificial Intelligence System for Detection and Diagnosis of Breast Lesion Using Mammography	2020-05-04	Yes	No	No		No
NCT03753724	IDEAL: Artificial Intelligence and Big Data for Early Lung Cancer Diagnosis Study	2022-03-31	Yes	No	No		No
NCT03851497	Application of Deep-learning and Ultrasound Elastography in Opportunistic Screening of Breast Cancer	2021-01-01	Yes	No	No		No
NCT04022512	Accuracy of Deep-learning Algorithm for Detection and Risk Stratification of Lung Nodules	2024-01-31	Yes	No	No		No
NCT04027556	Low-contrast Dose Liver CT Using Lean Body Weight Low Monoenergetic Images and Deep Learning-based Reconstruction	2022-02-10	Yes	No	Yes	36576547	Yes
NCT04036903	D-Lung: An Analytics Platform for Lung Cancer Based on Deep Learning Technology	2020-06-30	Yes	No	No		No
NCT04271657	RadioPathomics Artificial Intelligence Model to Predict nCRT Response in Locally Advanced Rectal Cancer	2020-12-30	Yes	No	Yes	34952679	Yes
NCT04317313	18F-FDG PET Radiomics of Diffuse Large B-cell Lymphoma	2021-07-30	Yes	Yes	Yes	34651227	Yes
NCT04340180	Evaluating the Performance of AI in Evaluating Breast MRI Performed With Dose Reduction	2026-01-20	No	No	No		No
NCT04441775	Artificial Intelligence for Prostate Cancer Treatment Planning	2022-03-04	Yes	No	No		No
NCT04688086	A Prospective Study to Evaluate Clinical Performance of Thermalix in Detecting Breast Cancers	2020-01-30	Yes	Yes	No		Yes
NCT04816981	AI-EBUS-Elastography for LN Staging	2022-05-01	Yes	No	No		No
NCT04838756	Mammography Screening With Artificial Intelligence (MASAI)	2025-08-12	No	No	Yes	41620232;39904652;37541274	Yes
NCT04893200	Radiomics-based Prediction Model of Tumor Spread Through Air Space in Lung Adenocarcinoma	2021-06-01	Yes	No	Yes	35529792	Yes
NCT04949776	Artificial Intelligence in Breast Cancer Screening Programs	2024-01-11	Yes	No	No		No
NCT04954378	Radiomics Based on Multimodal Imaging in Predicting Staging and Prognosis of Pancreatic Cancer	2020-12-30	Yes	No	No		No
NCT05048095	Artificial Intelligence in Breast Cancer Screening in Region Östergötland Linköping	2022-02-15	Yes	No	No		No
NCT05385718	Ezra Faster Scan Study	2025-02-24	Yes	No	No		No
NCT05481762	Evaluation of Contextflow DETECT Lung CT Nodule Detection Software in Chest CT Scans	2022-10-30	Yes	No	No		No
NCT05589480	Machine Learning for Recurrence Risk of Pancreatic Cancer After Radical Resection	2022-09-01	Yes	No	No		No
NCT05615298	Study to Evaluate Effectiveness of Investigational Device in the Assistance of Detection and Diagnosis of Breast Cancer	2021-01-13	Yes	No	No		No
NCT05640011	Clinical Validation of an Artificial Intelligence Algorithm to Help Interpret Mammograms	2023-05-20	Yes	No	No		No
NCT05723965	Using Artificial Intelligence to Predict Rectal Cancer Outcomes	2022-12-31	Yes	No	No		No
NCT05804799	Liver CT Dose Reduction With Deep Learning Based Reconstruction	2022-12-31	Yes	No	Yes	38231025	Yes

NCT identifier	Registered title	Completion date	Completed ≥12 months before extraction	Registry results posted	Empirical AI-results article linked by exact NCT	Linked PMID(s)	Any public result identified
NCT05820113	Deep Learning Super Resolution Reconstruction for Fast and Motion Robust T2-weighted Prostate MRI	2022-11-30	Yes	No	Yes	37750774	Yes
NCT05960188	Project 1: Self-Triage by 2D Full-field Digital Mammography or Synthetic Images	2023-03-04	Yes	No	No		No
NCT05979883	Randomized Evaluation of Machine Learning Assisted Radiation Treatment Planning vs Standard Radiation Treatment Planning	2025-10-29	No	No	No		No
NCT06037343	Deep Learning Image Reconstruction for Abdominal CT of Hepatocellular Carcinoma Compared With 3-TESLA MRI	2023-12-30	Yes	No	No		No
NCT06104189	Prospective Ultrafast MRI Radiomics for Breast Cancer	2023-08-31	Yes	No	Yes	39862364	Yes
NCT06168864	Development of Artificial Intelligence Models for Segmentation and Characterization of Prostate Cancer: a Single-center Retrospective Observational Study.	2022-06-01	Yes	No	No		No
NCT06187350	The Use of AI to Safely Reduce the Workload in Breast Cancer Screening With Mammography in Region Östergötland	2025-03-20	Yes	No	No		No
NCT06267690	A CT-based Radiomics Model to Predict Survival-graded Fibrosis in PDAC	2023-08-31	Yes	No	No		No
NCT06281769	Prostate Cancer Detection Rate of Targeted Biopsies With PCaVision	2025-08-15	No	No	No	41550796;39050912	No
NCT06306599	Human-AI Collaboration for Ultrasound Diagnosis of Thyroid Nodules - a Clinical Trial	2023-11-04	Yes	No	No		No
NCT06314750	Deep Learning Based MRI Radiomics in Predicting the Clinical Risk of Locally Advanced Rectal Cancer	2025-06-30	No	No	No		No
NCT06331975	Radiomics and Radiogenomics of Non-small Cell Lung Cancer	2023-09-14	Yes	No	No	42095156	No
NCT06336694	A Deep Learning Radiomics Model for Predicting Occult Peritoneal Metastases of Pancreatic Adenocarcinoma	2023-07-30	Yes	No	No		No
NCT06342934	Radiomics and Machine Learning in the Diagnosis of Ovarian Masses	2023-12-31	Yes	No	No		No
NCT06362291	Comparison of MRI AI-cTB Versus Routine cTB in Prostate Cancer Diagnosis: a Prospective Randomized Controlled Trial	2024-10-31	Yes	No	Yes	39533250	Yes
NCT06366906	10-year Retrospective Study of Oral and Maxillofacial Squamous Cell Carcinoma	2024-02-10	Yes	No	No		No
NCT06428799	Evaluation of Carebot AI MMG Medical Device for Breast Lesion Detection and Density Assessment	2024-04-24	Yes	No	No		No
NCT06479772	Evaluation of the Potential Impact of an Artificial Intelligence Solution in Second Reading of Organized Screening.	2024-03-31	Yes	No	No		No
NCT06521788	Artificial Intelligence Assisted Breast Ultrasound in Breast Cancer Screening	2023-12-31	Yes	No	Yes	41057952	Yes
NCT06546072	Deep Learning With MRI-based Multimodal-data Fusion Enhanced Postoperative Risk Stratification of Breast Cancer	2021-12-06	Yes	No	Yes	40345352	Yes
NCT06559046	A CT-BASED Deep Learning Model for Predicting WHO/ISUP Pathological Grades of Clear Cell Renal Cell Carcinoma (ccRCC) :A Multicenter Cohort Study	2024-06-30	Yes	No	No		No
NCT06575361	Comprehensive Evaluation of MRI-AI in Prostate Cancer Diagnosis	2025-08-31	No	No	No		No
NCT06576232	Standalone Observational Study Assessing the Performance of an AI/ML Tech-based SaMD on Chest LDCT Images (REALITY)	2024-08-21	Yes	No	No		No
NCT06589154	The Application of Multimodal Artificial Intelligence Systems in Prostate Cancer Diagnosis and Prognosis Analysis	2025-07-30	No	No	No		No
NCT06604078	Breast Cancer AI Imaging Study	2025-09-01	No	No	No		No
NCT06637059	Artificially Intelligent Model for Accurate Detection of HCC	2024-10-01	Yes	No	No		No
NCT06641947	Differentiation Benign and Malignant Pancreatic Lesions	2024-09-20	Yes	No	Yes	39832039	Yes

NCT identifier	Registered title	Completion date	Completed ≥12 months before extraction	Registry results posted	Empirical AI-results article linked by exact NCT	Linked PMID(s)	Any public result identified
NCT06659601	Deep Learning Model to Predict the Recurrence of Stage IA Invasive Lung Adenocarcinoma After Sub-lobar Resection	2024-10-24	Yes	No	No		No
NCT06737367	Integrating Machine Learning for Prognostic Prediction in Stage I NSCLC by CT Images and Pathological Factors	2024-11-11	Yes	No	No		No
NCT06757530	Radiomics Model for Assessing Lymph Node Status in cN0 Patients withHNSCC	2025-04-15	Yes	No	No		No
NCT06761742	Development of AI Model for Renal Tumor Diagnosis Using CT and Lab Tests	2024-12-01	Yes	No	No		No
NCT06853457	A Transfer Learning Radiomics Model for Predicting Response to Initial Transarterial Embolization in Patients with Gastroenteropancreatic Neuroendocrine Tumor Liver Metastases	2024-09-01	Yes	No	No		No
NCT06885333	Whole-Brain Tumor Burden Metrics with Inflammatory and Molecular Markers to Predict Postoperative Neurocognitive Decline in Glioma	2024-12-01	Yes	No	No		No
NCT06888089	Deep Clinical Trajectory Modeling to Optimize Accrual to Cancer Clinical Trials	2024-07-15	Yes	No	Yes	40257799	Yes
NCT06984289	US Radiomics in Advanced Cervical Cancer	2025-05-14	No	No	No		No
NCT07065422	AI Models for Predicting Occult Pleural Dissemination in NSCLC	2025-01-01	Yes	No	No		No
NCT07117786	Computed Tomography Radiomics-Derived Nomogram for Predicting Early Renal Function Decline After Partial Nephrectomy in Renal Cell Carcinoma: A Multicenter Development/Validation Study	2023-06-01	Yes	No	No		No
NCT07153783	Interventional AI-Human Collaboration for Liver Tumor Diagnosis	2025-11-07	No	No	No		No
NCT07181954	CT-based Radiomics for Prediction of MIT Family Translocation Kidney Cancer: a Multicenter, Retrospective Clinical Study	2023-12-31	Yes	No	No		No
NCT07190040	Integrating Multi-Omics Data for Enhanced Prognosis Prediction in Gastric Cancer Post-Neoadjuvant Therapy	2025-09-01	No	No	No		No
NCT07251946	Development of Multiphasic CT-based Delta-Radiomics Model for Predicting Postoperative Recurrence Risk in Bladder Cancer	2023-03-31	Yes	No	No		No
NCT07287111	Effectiveness of CadAI-B Dx for Decision Support in Breast Ultrasound	2025-11-24	No	No	No		No
NCT07350382	Predicting Cognitive Dysfunction After Pediatric Posterior Fossa Tumor Surgery	2024-01-01	Yes	No	No		No
NCT07496684	Application and Evaluation of Vision-LSTM Modeling in Diagnostic Ultrasound Imaging of TI-RADS Category 4b Thyroid Nodules	2024-12-30	Yes	No	No		No

Note: Among 72 completed studies, 60 had been completed at least 12 months before extraction; 3 had posted registry results, 13 had an empirical AI-results PubMed article linked by exact NCT identifier, and 15 had either source. Blank PMID cells indicate no qualifying exact-NCT link.